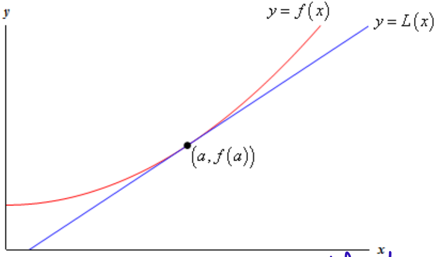
We will cover the following five topics in the Week 3 lectures:

1. Tangent plane functions.
2. Linear approximations and total differentiability. textbook omits the descriptor "total"
3. The Multivariable Chain Rule.
4. The Implicit Function Theorem and implicit differentiation.
5. Directional derivatives.

Tangent Line Functions and Linear Approximations



continuous

Let f be a real-valued function of one real variable ~~with an open domain~~ defined on an open interval.

The graph of f, which is the subset of $\mathbb{R}^2$ described by

$$\left\{(x, y) \in \mathbb{R}^2 \;\middle|\; x \in \mathrm{Dom}(f) \text{ and } y = f(x)\right\},$$ is a curve in $\mathbb{R}^2$.

can be given the so-called "standard" parametrization $\sigma_f : \mathrm{Dom}(f) \to \mathbb{R}^2$, defined by

vector function

$$\forall x \in \underbrace{\mathrm{Dom}(f)}_{\text{interval}} : \quad \boldsymbol{\sigma}_f(x) \stackrel{\text{df}}{=} (x, f(x)).$$

If f is differentiable at a, then the following statements hold: $\frac{d}{dx}[(x,f(x))]\big|_{x=a} = (1, f'(a))$

- sigma $\boldsymbol{\sigma}_f$ is differentiable at a, and the tangent vector of $\boldsymbol{\sigma}_f$ at a is $(1, f'(a))$.
- The tangent line to the graph of f at the point $(a, f(a))$ induced by $\boldsymbol{\sigma}_f$ possesses the so-called "standard" affine parametrization $\mathbf{r}_f : \mathbb{R} \to \mathbb{R}^2$, defined by

for this parameterization, we want the first coordinate to be the parameter

$$\forall x \in \mathbb{R} : \quad \mathbf{r}_f(x) \overset{\text{df}}{=} (x, f(a) + f'(a) \cdot (x - a)).$$

$\vec{r}_f(a) = (a, f(a))$.

This motivates the following definition of tangent line functions.

Definition (Tangent Line Functions)

- Let f be a real-valued function of one real variable with an open domain.
- Let $a \in \text{Dom}(f)$.

If f is differentiable at a, then the tangent line function of f at a is the linear function $L_{f,a} : \mathbb{R} \to \mathbb{R}$, defined by

$$L_{f,a}(x) \overset{\text{df}}{=} f(a) + f'(a) \cdot (x - a).$$

Definition (Linear Approximations)

- Let f be a real-valued function of one real variable with an open domain.
- Let $a \in \mathrm{Dom}(f)$.

$y = mx + c$

A linear approximation of f at a is defined as a linear function $L : \mathbb{R} \to \mathbb{R}$ such that

- $L(a) = f(a)$ and
- $\lim_{x \to a} \dfrac{|f(x) - L(x)|}{|x - a|} = 0$.

In Calc I, existence of linear approx. and differentiability are the same

In Calculus I, you may have seen the following theorem.

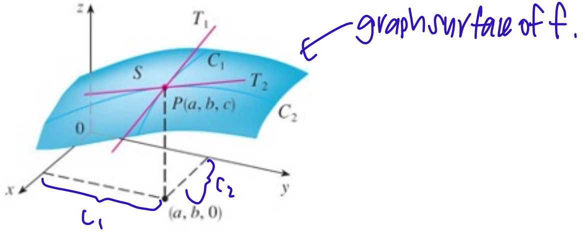
Theorem (Differentiability and the Existence of a Linear Approximation)

- Let f be a real-valued function of one real variable with an open domain.
- Let $a \in \mathrm{Dom}(f)$.

Then we can find a linear approximation of f at a if and only if f is differentiable at a, in which case the linear approximation is precisely $L_{f,a}$.

Therefore, the existence of a linear approximation is equivalent to differentiability.

In Calculus III, we want to generalize the concept of tangent line functions to get the concept of tangent plane functions.



Let f be a real-valued function of two real variables with an open domain.

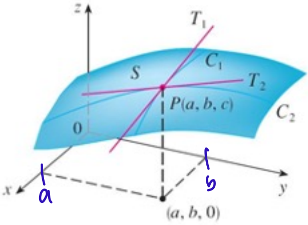
Let $(a, b) \in \mathrm{Dom}(f)$, let $c \overset{\text{df}}{=} f(a, b)$, and let $P \overset{\text{df}}{=} (a, b, c)$.

Suppose that the x- and y-partial derivatives of f at (a, b) exist.

The vertical trace of f made by the $y = b$ plane is denoted by C_1.

The vertical trace of f made by the $x = a$ plane is denoted by C_2.

D_1 & D_2 are domains we want to fix



Let $D_1 \stackrel{\text{df}}{=} \{t \in \mathbb{R} \mid (t, b) \in \text{Dom}(f)\}$, which is guaranteed to be an open subset of $\mathbb{R}$.

fix $y = b$

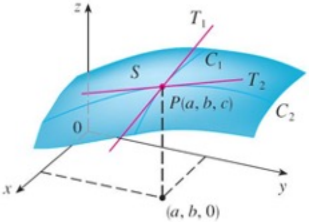
C_1 is parametrized by the function $\mathbf{v}_1 : D_1 \to \mathbb{R}^3$, defined by $\mathbf{v}_1(t) \stackrel{\text{df}}{=} (t, b, f(t, b))$.

Let $D_2 \stackrel{\text{df}}{=} \{t \in \mathbb{R} \mid (a, t) \in \text{Dom}(f)\}$, which is guaranteed to be an open subset of $\mathbb{R}$.

fix $x = a$

C_2 is parametrized by the function $\mathbf{v}_2 : D_2 \to \mathbb{R}^3$, defined by $\mathbf{v}_2(t) \stackrel{\text{df}}{=} (a, t, f(a, t))$.

eventually will differentiate w.r.t t, which is the parameter



The tangent vector to C_1 at P induced by $\mathbf{v}_1$ is

$$\mathbf{v}_1'(a) = \frac{\mathrm{d}}{\mathrm{d}t}[(t, b, f(t, b))]|_{t=a} = (\overset{\neq 0}{1}, 0, f_x(a, b)) \neq \mathbf{0}.$$

The tangent vector to C_2 at P induced by $\mathbf{v}_2$ is

$$\mathbf{v}_2'(b) = \frac{\mathrm{d}}{\mathrm{d}t}[(a, t, f(a, t))]|_{t=b} = (0, \overset{\neq 0}{1}, f_y(a, b)) \neq \mathbf{0}.$$

non-zero tangent vectors
$\Downarrow$
they are not scalar multiples of each other
also not parallel

We want to find a plane Π in $\mathbb{R}^3$ that meets the following two requirements:

- Π contains P.
- Π is parallel to the non-parallel non-zero vectors $\mathbf{v}_1'(a)$ and $\mathbf{v}_2'(b)$.

Recall that the scalar product $\bullet$ for vectors in $\mathbb{R}^3$ takes two vectors in $\mathbb{R}^3$ and returns a real scalar.

More precisely, for vectors **a** and **b** in $\mathbb{R}^3$, we have

$$\mathbf{a} \bullet \mathbf{b} = \begin{cases} \|\mathbf{a}\|_2 \|\mathbf{b}\|_2 \cos(\theta_{\mathbf{a},\mathbf{b}}), & \text{if } \mathbf{a}, \mathbf{b} \neq \mathbf{0}; \\ 0, & \text{if } \mathbf{a} = \mathbf{0} \text{ or } \mathbf{b} = \mathbf{0}, \end{cases}$$

where $\theta_{\mathbf{a},\mathbf{b}}$ denotes the unique angle between **a** and **b** with value in $[0, \pi]$.

3only

There is a different product for vectors in $\mathbb{R}^3$, known as the cross product or the vector product, and denoted by "$\times$", that takes two vectors in $\mathbb{R}^3$ and returns a vector in $\mathbb{R}^3$.

Definition (the Cross/Vector Product)

Let $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ be vectors in $\mathbb{R}^3$.

The cross/vector product of **a** with **b**, denoted by "$\mathbf{a} \times \mathbf{b}$", is defined as

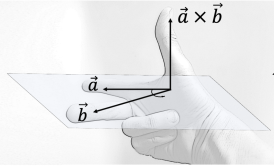
$$\mathbf{a} \times \mathbf{b} \stackrel{\text{df}}{=} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2) \cdot \mathbf{i} - (a_1 b_3 - a_3 b_1) \cdot \mathbf{j} + (a_1 b_2 - a_2 b_1) \cdot \mathbf{k}.$$

Theorem (a Geometric Characterization of Cross/Vector Product)

For vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$, we have

$$\mathbf{a} \times \mathbf{b} = \begin{cases} \|\mathbf{a}\|_2 \|\mathbf{b}\|_2 \sin(\theta_{\mathbf{a},\mathbf{b}}) \cdot \mathbf{n}_{\mathbf{a},\mathbf{b}}, & \text{if } \mathbf{a}, \mathbf{b} \neq \mathbf{0} \text{ and } \mathbf{a} \nparallel \mathbf{b}; \\ \mathbf{0}, & \text{if } \mathbf{a} = \mathbf{0},\ \mathbf{b} = \mathbf{0},\ \text{or } (\mathbf{a}, \mathbf{b} \neq \mathbf{0} \text{ and } \mathbf{a} \parallel \mathbf{b}), \end{cases}$$

where $\mathbf{n}_{\mathbf{a},\mathbf{b}}$ denotes the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ such that $(\mathbf{a}, \mathbf{b}, \mathbf{n}_{\mathbf{a},\mathbf{b}})$ has the same orientation as $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ (physically represented by the Right-Hand Rule).



In short, the cross/vector product of two non-parallel non-zero vectors in $\mathbb{R}^3$ is a vector in $\mathbb{R}^3$ that is perpendicular to the former two.

Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^3$. Let c be a real scalar.

Then the following statements hold:

1. Anti-commutativity: $\mathbf{x} \times \mathbf{y} = -(\mathbf{y} \times \mathbf{x})$. in general
2. Left distributivity over vector addition: $\mathbf{x} \times (\mathbf{y} + \mathbf{z}) = (\mathbf{x} \times \mathbf{y}) + (\mathbf{x} \times \mathbf{z})$.
3. Compatibility with scalar multiplication: $c \cdot (\mathbf{x} \times \mathbf{y}) = (c \cdot \mathbf{x}) \times \mathbf{y} = \mathbf{x} \times (c \cdot \mathbf{y})$.

$\vec{x} \times \vec{x} = -(\vec{x} \times \vec{x})$

$(\vec{x} \times \vec{x}) + (\vec{x} \times \vec{x}) = \vec{0}$

$2(\vec{x} \times \vec{x}) = \vec{0}$

$\therefore \vec{x} \times \vec{x} = \vec{0}$

It follows from anti-commutativity that $\mathbf{x} \times \mathbf{x} = \mathbf{0}$.

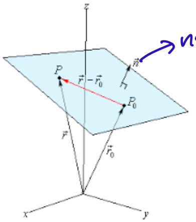
It follows from left distributivity over vector addition that $\mathbf{x} \times \mathbf{0} = \mathbf{0}$.

Right distributivity over vector addition follows from both left distributivity over vector addition and anti-commutativity.

ASSOCIATIVITY DOES NOT HOLD!

$\vec{x} \times (\vec{y} \times \vec{z}) \neq (\vec{x} \times \vec{y}) \times \vec{z}$

in general



normal vector

$ax + by + cz = d$

$\vec{n} = (a, b, c)$

Theorem (Plane Existence Theorem 1)

- Let $P_0 = (x_0, y_0, z_0) \in \mathbb{R}^3$.
- Let $\mathbf{n} = (A, B, C)$ be a non-zero vector in $\mathbb{R}^3$.

Then there exists a unique plane in $\mathbb{R}^3$ that contains P_0 and is perpendicular to $\mathbf{n}$; it is described explicitly as the set of all $P = (x, y, z) \in \mathbb{R}^3$ such that $\mathbf{n} \perp \overrightarrow{P_0P}$:

$$\begin{aligned} 0 &= \mathbf{n} \bullet \overrightarrow{P_0P} \\ &= (A, B, C) \bullet (x - x_0, y - y_0, z - z_0) \\ &= A(x - x_0) + B(y - y_0) + C(z - z_0). \end{aligned}$$

Theorem (Plane Existence Theorem 2)

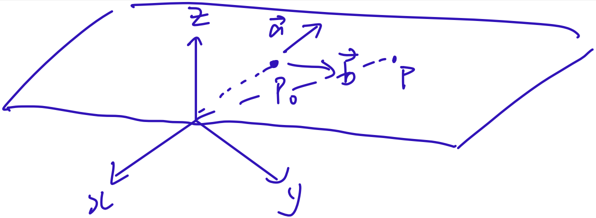
- Let $P_0 = (x_0, y_0, z_0) \in \mathbb{R}^3$.
- Let $\mathbf{a}$ and $\mathbf{b}$ be non-parallel non-zero vectors in $\mathbb{R}^3$, and let $\mathbf{n} \overset{\text{df}}{=} \mathbf{a} \times \mathbf{b}$.

Then there exists a unique plane in $\mathbb{R}^3$ that contains P_0 and is parallel to $\mathbf{a}$ and $\mathbf{b}$; it is described explicitly as the set

→ vector eqn of a plane (JC)

$$\left\{ P \in \mathbb{R}^3 \;\middle|\; \exists \lambda, \mu \in \mathbb{R} : \quad \overrightarrow{OP} = \overrightarrow{OP_0} + \lambda \cdot \mathbf{a} + \mu \cdot \mathbf{b} \right\},$$

and it coincides with the plane in $\mathbb{R}^3$ that contains P_0 and is perpendicular to $\mathbf{n}$.



Back to Tangent Plane Functions

Now, let $\mathbf{n} \overset{\text{df}}{=} \mathbf{v}_1'(a) \times \mathbf{v}_2'(b)$; explicitly,

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_x(a,b) \\ 0 & 1 & f_y(a,b) \end{vmatrix} = \underline{(-f_x(a,b), -f_y(a,b), 1)} \neq \mathbf{0}.$$

By Plane Existence Theorem 2, the following two planes in $\mathbb{R}^3$ are the same:

- The plane in $\mathbb{R}^3$ that contains P and is parallel to $\mathbf{v}_1'(a)$ and $\mathbf{v}_2'(b)$.
- The plane in $\mathbb{R}^3$ that contains P and is perpendicular to $\mathbf{n}$.

This plane is described explicitly as the set of all $Q = (x, y, z) \in \mathbb{R}^3$ such that

$$\begin{aligned}
0 &= \mathbf{n} \bullet \cancel{\overrightarrow{PQ}}\ \overrightarrow{P_0P} \\
&= (-f_x(a,b), -f_y(a,b), 1) \bullet (x-a, y-b, z-c) \\
&= (-f_x(a,b), -f_y(a,b), 1) \bullet (x-a, y-b, z-f(a,b)) \\
&= -f_x(a,b)\cdot(x-a) - f_y(a,b)\cdot(y-b) + (z - f(a,b)); \\
z &= f(a,b) + f_x(a,b)\cdot(x-a) + f_y(a,b)\cdot(y-b).
\end{aligned}$$

tangent plane function

Definition (Tangent Plane Functions)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(f)$.

If the x- and y-partial derivatives of f at $\mathbf{p}$ exist, then the tangent plane function of f at $\mathbf{p}$ is the function $L_{f,\mathbf{p}} : \mathbb{R}^2 \to \mathbb{R}$, defined by

$$\begin{aligned} \forall \mathbf{x} = (x, y) \in \mathbb{R}^2 : \quad L_{f,\mathbf{p}}(\mathbf{x}) &= L_{f,\mathbf{p}}(x, y) \\ &\overset{\text{df}}{=} f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b) \\ &= f(\mathbf{p}) + (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet (\mathbf{x} - \mathbf{p}). \end{aligned}$$

Question: In single-variable calculus, tangent line functions and linear approximations coincide. In multivariable calculus, should we expect tangent plane functions and linear (or planar) approximations to coincide as well? $\Rightarrow$ No. See later on.

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} 2x^2 + y^2$.
Problem: Find the tangent-plane function of f at $(1, 1)$.

Solution

differentiate wrt x differentiate wrt y

- Given (x, y) in $\mathbb{R}^2$, we have $f_x(x, y) = 4x$ and $f_y(x, y) = 2y$.
- It follows that $f_x(1, 1) = 4$ and $f_y(1, 1) = 2$.
- The tangent plane function of f at $(1, 1)$, denoted by $L_{f,(1,1)}$, is therefore given by

$$\forall x, y \in \mathbb{R} : \quad \begin{aligned} L_{f,(1,1)}(x, y) &= f(1, 1) + f_x(1, 1) \cdot (x - 1) + f_y(1, 1) \cdot (y - 1) \\ &= 3 + 4(x - 1) + 2(y - 1) \\ &= 4x + 2y - 3. \end{aligned}$$

formula on previous slide

Total Differentiability — a Definition Based on Linear Approximations

1D: $f(x) = mx + c$

Definition (Linear Functions of Two Real Variables)

A linear/planar function of two real variables is defined as a function $f : \mathbb{R}^2 \to \mathbb{R}$ with the property that there exist $a, b, c \in \mathbb{R}$ such that

$$\forall x, y \in \mathbb{R} : \quad f(x, y) = ax + by + c.$$

Definition (Linear Approximations and Total Differentiability)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} \in \mathrm{Dom}(f)$.

A linear approximation of f at $\mathbf{p}$ is defined as a linear/planar function L such that

- $L(\mathbf{p}) = f(\mathbf{p})$ and
- $\displaystyle \lim_{\mathbf{x} \to \mathbf{p}} \frac{|f(\mathbf{x}) - L(\mathbf{x})|}{\|\mathbf{x} - \mathbf{p}\|_2} = 0.$

Note: A linear approximation of f at $\mathbf{p}$ may not exist, but if it exists, then it is unique!

We say that f is totally differentiable at $\mathbf{p}$ if and only if a linear approximation of f at $\mathbf{p}$ exists.

Theorem (Total Differentiability and the Existence of Partial Derivatives)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(f)$.

If f is totally differentiable at $\mathbf{p}$, then the following statements hold:

- The x- and y-partial derivatives of f at $\mathbf{p}$ exist.
- The linear approximation L of f at $\mathbf{p}$ is the tangent plane function of f at $\mathbf{p}$:

$$\forall \mathbf{x} \in \mathbb{R}^2 : \quad L(\mathbf{x}) = L_{f,\mathbf{p}}(\mathbf{x}) = f(\mathbf{p}) + (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet (\mathbf{x} - \mathbf{p}).$$

Caution: In this definition, it is nowhere stated that if the x- and y-partial derivatives of f at $\mathbf{p}$ exist, then the tangent plane function of f at $\mathbf{p}$ is a linear approximation of f at $\mathbf{p}$. THIS STATEMENT IS FALSE IN GENERAL!

Summary: The existence of a linear approximation at a point is a stronger requirement than the existence of partial derivatives at the same point.

Theorem (Total Differentiability Implies Continuity)

Let f be a real-valued function of two real variables with an open domain.

If f is totally differentiable at $\mathbf{p} \in \text{Dom}(f)$, then f is continuous at $\mathbf{p}$.

This theorem gives rise to the following useful corollary. contrapositive

Corollary (Discontinuity Implies Non-Total-Differentiability)

Let f be a real-valued function of two real variables with an open domain.

If f is <u>not</u> continuous at $\mathbf{p} \in \text{Dom}(f)$, then f is <u>not</u> totally differentiable at $\mathbf{p}$.

Definition (an Alternative Definition of Total Differentiability)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(f)$.

Then f is said to be totally differentiable at $\mathbf{p}$ if and only if there exist

(i) real numbers A and B, and

(ii) functions $\epsilon_1 : \mathrm{Dom}(f) \to \mathbb{R}$ and $\epsilon_2 : \mathrm{Dom}(f) \to \mathbb{R}$,

$\varepsilon_1, \varepsilon_2$ are real valued functions with domain $\mathrm{Dom}(f)$

with the following two properties:

- $\lim_{\mathbf{x} \to \mathbf{p}} \epsilon_1(\mathbf{x}) = \lim_{\mathbf{x} \to \mathbf{p}} \epsilon_2(\mathbf{x}) = 0$.
- Given $(x, y) \in \mathrm{Dom}(f)$, we have

$$f(x,y) - f(a,b) = A\underbrace{(x-a)}_{\Delta x} \pm B\underbrace{(y-b)}_{\Delta y} \pm \epsilon_1(x,y)\cdot\underbrace{(x-a)}_{\Delta x} \pm \epsilon_2(x,y)\cdot\underbrace{(y-b)}_{\Delta y}.$$

Remarks:

- The first definition is suited to refuting total differentiability.
- The alternative definition is suited to establishing total differentiability.

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x,y) \overset{\text{df}}{=} 2x^2 - xy$.
Problem: Show that f is totally differentiable throughout its domain.

show @ arbitrary point P

Solution

- Fix $\mathbf{p} = (a,b) \in \mathbb{R}^2$, and let $(x,y) \in \mathbb{R}^2$.
- Let $\Delta x \overset{\text{df}}{=} x - a$, so that $x = a + \Delta x$. change in x
- Let $\Delta y \overset{\text{df}}{=} y - b$, so that $y = b + \Delta y$. change in y
- Then

$f_x(x,y) = 4x - y$
$f_y(x,y) = -x$
$\Downarrow$
based on Theorem on slide 22, this is an easier and shorter method

$$\begin{aligned} f(x,y) - f(a,b) &= f(a+\Delta x, b+\Delta y) - f(a,b) \\ &= \Big(2(a+\Delta x)^2 - (a+\Delta x)(b+\Delta y)\Big) - \Big(2a^2 - ab\Big) \\ &= \underbrace{(4a-b)}_{A}(\Delta x) - \underbrace{a}_{B}(\Delta y) + \underbrace{2(\Delta x)}_{\varepsilon_1}(\Delta x) - \underbrace{(\Delta x)}_{\varepsilon_2}(\Delta y). \end{aligned}$$

$(x,y) \to (a,b)$

- Let $A \overset{\text{df}}{=} 4a - b$ and $B \overset{\text{df}}{=} a$.
- Define functions $\epsilon_1, \epsilon_2 : \text{Dom}(f) \to \mathbb{R}^2$ by

$$\epsilon_1(x, y) \overset{\text{df}}{=} 2(\underbrace{\overset{\to 0}{x - a}}_{\Delta x}) \quad \text{and} \quad \epsilon_2(x, y) \overset{\text{df}}{=} \overset{\to 0}{x - a}.$$

we let $(x,y) \to (a,b)$

- Then given $(x, y) \in \text{Dom}(f)$, we have

$$f(x, y) - f(a, b) = A(x - a) - B(y - b) + \epsilon_1(x, y) \cdot (x - a) - \epsilon_2(x, y) \cdot (y - b).$$

- Furthermore, $\lim_{\mathbf{x} \to \mathbf{p}} \epsilon_1(\mathbf{x}) = \lim_{\mathbf{x} \to \mathbf{p}} \epsilon_2(\mathbf{x}) = 0$.
- Hence, f is totally differentiable at $\mathbf{p}$. $= (a,b)$
- Finally, as $\mathbf{p}$ is an arbitrary point in $\mathbb{R}^2$, we conclude that f is totally differentiable throughout its domain.

Theorem (a Sufficient Criterion for Total Differentiability)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} \in \mathrm{Dom}(f)$.

If there exists an open disk D in $\mathbb{R}^2$, centered at $\mathbf{p}$ and contained within $\mathrm{Dom}(f)$, such that f_x and f_y are defined and continuous throughout D, then f is totally differentiable at $\mathbf{p}$.

Had we known about this theorem earlier, the previous two slides would not have been so tortuous.

If you are, however, thinking that the existence of partial derivatives at a point implies total differentiability at the same point, then think again.

total diff $\Rightarrow$ existence of partial derivatives

not true the other way!

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) \overset{\text{df}}{=} \begin{cases} \dfrac{xy}{x^2+y^2}, & \text{if } (x,y) \neq (0,0); \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Evaluate $f_x(0,0)$ and $f_y(0,0)$, if they exist.

(b) Show that f is <u>not</u> totally differentiable at $(0,0)$.

Solution

fix $y=0$

(a)
- For all $x \in \mathbb{R}$, we have $f(x,0) = 0$.
- Hence, $f_x(0,0) = \dfrac{\mathrm{d}}{\mathrm{d}x}[f(x,0)]|_{x=0} = \dfrac{\mathrm{d}}{\mathrm{d}x}[0]|_{x=0} = 0|_{x=0} = 0.$
- For all $y \in \mathbb{R}$, we have $f(0,y) = 0$. → fix $x=0$
- Hence, $f_y(0,0) = \dfrac{\mathrm{d}}{\mathrm{d}y}[f(0,y)]|_{y=0} = \dfrac{\mathrm{d}}{\mathrm{d}y}[0]|_{y=0} = 0|_{y=0} = 0.$

Total Differentiability — Example 2 (cont'd)

(b)

- For the sake of contradiction, suppose that f is totally differentiable at $(0,0)$.
- The tangent plane function of f at $(0,0)$, denoted by $L_{f,(0,0)}$, is given by

differentiate wrt x

$$\forall x, y \in \mathbb{R}: \quad L_{f,(0,0)}(x,y) = f(0,0) + f_x(0,0)\cdot(x-0) + f_y(0,0)\cdot(y-0)$$
$$= 0 + 0(x-0) + 0(y-0)$$
$$= 0.$$

- By the first definition of total differentiability, we then have

linear approximation

$$\lim_{(x,y)\to(0,0)} \frac{\left|f(x,y) - L_{f,(0,0)}(x,y)\right|}{\|(x,y)-(0,0)\|_2} = 0;$$

origin is the point of interest here

$\|(x,y)\| = \sqrt{x^2+y^2}$

$$\lim_{(x,y)\to(0,0)} \frac{\left(\frac{|xy|}{x^2+y^2}\right)}{\sqrt{x^2+y^2}} = 0;$$

$$\lim_{(x,y)\to(0,0)} \frac{|xy|}{\sqrt{(x^2+y^2)^3}} = 0.$$

- Restrict the limit in the preceding equation to the line given by $y = x$:

contradict!

$$\lim_{x\to 0} \frac{|x|^2}{\sqrt{8x^6}} = \lim_{x\to 0} \frac{|x|^2}{\sqrt{8}|x|^3} = \lim_{x\to 0} \frac{1}{\sqrt{8}|x|} = \infty. \quad \text{(Contradiction)}$$

- Therefore, f is not totally differentiable at $(0,0)$.

Total Differentiability — Linear Approximations

Theorem (the Linear-Approximation Formula)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(f)$.
- Suppose that f is totally differentiable at $\mathbf{p}$. → there is a linear approx.

Then for all $\mathbf{x} \in \mathrm{Dom}(f)$ sufficiently close to $\mathbf{p}$, we have $f(\mathbf{x}) \approx L_{f,\mathbf{p}}(\mathbf{x})$, which yields

$$\underbrace{f(\mathbf{x}) - f(\mathbf{p})}_{\Delta z} \approx f_x(\mathbf{p}) \cdot \underbrace{(x-a)}_{\Delta x} + f_y(\mathbf{p}) \cdot \underbrace{(y-b)}_{\Delta y}.$$

change in output

change in input

tangent plane formula is a linear approx

This theorem is a little vague because — as with all approximation results — it should be accompanied by error bounds.

We will, however, apply it in its current state.

no total differentiability $\Rightarrow$ no linear approx.

Example

- The base radius and height of a cone are measured to be 10 and 25, respectively.
- The absolute error in each measurement is at most 0.1.

Problem: Use a linear approximation to estimate the maximum absolute error involved in calculating the volume of the cone from these measurements.

Solution

- The volume V, base radius r, and height h of a cone are related as $V = \frac{1}{3}\pi r^2 h$.
- We thus define a function $V : \mathbb{R}_{>0} \times \mathbb{R}_{>0} \to \mathbb{R}$ by $V(r, h) \overset{\text{df}}{=} \frac{1}{3}\pi r^2 h$.
- Let us first show that V is totally differentiable throughout its domain. → must show this first! to use linear approx.
- Note that $\frac{\partial V}{\partial r} = \frac{2}{3}\pi rh$ and $\frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2$. ⇒ continuous functions
- These expressions yield continuous functions from $\mathbb{R}_{>0} \times \mathbb{R}_{>0}$ to $\mathbb{R}$.
- Hence, V is totally differentiable throughout its domain, particularly at $(10, 25)$.
- Therefore, the tangent plane function of V at $(10, 25)$ is a linear approximation of V at $(10, 25)$, and the requested maximum absolute error is

sub into $\frac{2}{3}\pi rh$

$$|\Delta V| \approx \underline{\frac{\partial V}{\partial r}}(10, 25) \cdot \underset{0.1}{|\Delta r|} + \frac{\partial V}{\partial h}(10, 25) \cdot \underset{0.1}{|\Delta h|} = \frac{500\pi}{3}(0.1) + \frac{100\pi}{3}(0.1) = 20\pi.$$

Recall the Chain Rule for single-variable calculus.

Theorem (the Single-Variable Chain Rule)

- Let f and g be real-valued functions of one real variable with an open domain.
- Let $a \in \operatorname{Dom}(f)$ satisfy $f(a) \in \operatorname{Dom}(g)$.

If

- f is differentiable at a, and
- g is differentiable at $f(a)$,

then $g \circ f$ is differentiable at a, and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a).$$

Symbolically, in the Leibniz Notation, we write this as $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \dfrac{\mathrm{d}y}{\mathrm{d}x}\dfrac{\mathrm{d}x}{\mathrm{d}t}$.

independent variable

intermediate variable

Theorem (the Multivariable Chain Rule (the $(2, 1)$ Version))

- Let f be a real-valued function of two real variables with an open domain.
- Let g and h be real-valued functions of one real variable with an open domain.
- Let $a \in \mathrm{Dom}(g) \cap \mathrm{Dom}(h)$ satisfy $(g(a), h(a)) \in \mathrm{Dom}(f)$.

$f(g(x), h(x))$

If

- f is totally differentiable at $(g(a), h(a))$, and
- g and h are differentiable at a,

then $f \circ (g, h)$ is differentiable at a, and

$$
\begin{aligned}
(f \circ (g, h))'(a) &= \frac{\mathrm{d}}{\mathrm{d}x}[f(g(x), h(x))]|_{x=a} \\
&= f_x(g(a), h(a)) \cdot g'(a) + f_y(g(a), h(a)) \cdot h'(a).
\end{aligned}
$$

Symbolically, in the Leibniz Notation, we write this as $\dfrac{\mathrm{d}z}{\mathrm{d}t} = \dfrac{\partial z}{\partial x}\dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{\partial z}{\partial y}\dfrac{\mathrm{d}y}{\mathrm{d}t}$.

Example

- Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x,y) \overset{\text{df}}{=} x^2 e^y$. f is the outer function.
- Define $g : \mathbb{R} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ by $g(t) \overset{\text{df}}{=} t^2 - 1$ and $h(t) \overset{\text{df}}{=} \sin(t)$. g and h are inner func.
- Define $F : \mathbb{R} \to \mathbb{R}$ by $F(t) \overset{\text{df}}{=} f(g(t), h(t))$. $f \circ (g,h)$

Problem: Use the $(2,1)$ version of the Multivariable Chain Rule to determine F'.

Solution

- For all $x, y \in \mathbb{R}$, we have $f_x(x,y) = 2xe^y$ and $f_y(x,y) = x^2 e^y$. differentiate outer func. first
- As f_x and f_y are defined and continuous throughout $\mathbb{R}^2$, it follows that f is totally differentiable throughout $\mathbb{R}^2$.
- For all $t \in \mathbb{R}$, we have $g'(t) = 2t$ and $h'(t) = \cos(t)$.
- Then the $(2,1)$ version of the Multivariable Chain Rule says that

$$\begin{aligned}
\forall t \in \mathbb{R} : \quad F'(t) &= (f \circ (g,h))'(t) \\
&= f_x(g(t), h(t)) \cdot g'(t) + f_y(g(t), h(t)) \cdot h'(t) \\
&= 2g(t)e^{h(t)} \cdot g'(t) + [g(t)]^2 e^{h(t)} \cdot h'(t) \\
&= 4t\left(t^2 - 1\right)e^{\sin(t)} + \left(t^2 - 1\right)^2 e^{\sin(t)} \cos(t).
\end{aligned}$$

outer func. inner func.

Theorem (the Multivariable Chain Rule (the $(2,2)$ Version))

- Let f be a real-valued function of two real variables with an open domain.
- Let g and h be real-valued functions of two real variables with an open domain.
- Let $\mathbf{p} \in \mathrm{Dom}(g) \cap \mathrm{Dom}(h)$ satisfy $(g(\mathbf{p}), h(\mathbf{p})) \in \mathrm{Dom}(f)$.

If

- f is totally differentiable at $(g(\mathbf{p}), h(\mathbf{p}))$, and
- g and h are totally differentiable at $\mathbf{p}$,

then $f \circ (g, h)$ is totally differentiable at $\mathbf{p}$, and

we have 2 independent variables, so 2 derivatives.

$$\begin{aligned}(f \circ (g,h))_x(\mathbf{p}) &= \frac{\partial}{\partial x}[f(g(\mathbf{x}), h(\mathbf{x}))]|_{\mathbf{x}=\mathbf{p}} \\ &= f_x(g(\mathbf{p}), h(\mathbf{p})) \cdot g_x(\mathbf{p}) + f_y(g(\mathbf{p}), h(\mathbf{p})) \cdot h_x(\mathbf{p});\end{aligned}$$

$\Rightarrow$ x independen variable

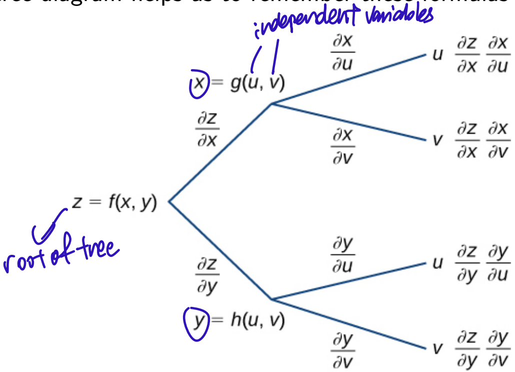
$$\begin{aligned}(f \circ (g,h))_y(\mathbf{p}) &= \frac{\partial}{\partial y}[f(g(\mathbf{x}), h(\mathbf{x}))]|_{\mathbf{x}=\mathbf{p}} \\ &= f_x(g(\mathbf{p}), h(\mathbf{p})) \cdot g_y(\mathbf{p}) + f_y(g(\mathbf{p}), h(\mathbf{p})) \cdot h_y(\mathbf{p}).\end{aligned}$$

$\Rightarrow$ y independent variable

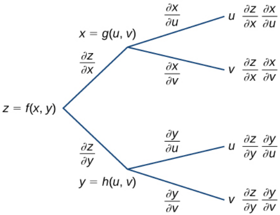
The $(2,2)$ version of the Multivariable Chain Rule can be written symbolically as

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u};$$
$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}.$$

The following tree diagram helps us to remember these formulas more easily.



The Multivariable Chain Rule (the $(2,2)$ Version) — a Mnemonic



- From z, draw branches to the intermediate variables, x and y.
- From x and from y, draw branches to the independent variables, u and v.
- Every branch represents the partial derivative of its source variable with respect to its target variable.
- To write the Chain Rule, first select an independent variable (u or v).
- For every path from z to the selected independent variable, multiply all the partial derivatives represented by the branches belonging to the path.
- Finally, add all these products of partial derivatives to get the Chain Rule formula for the selected independent variable.

↳ will hardly apply in this course... YAY!

Theorem (the Multivariable Chain Rule (the (m, n) Version))

- Let f be a real-valued function of m real variables with an open domain.
- Let $g_1, \ldots, g_m$ be real-valued functions of n real variables with an open domain.
- Let $\mathbf{p} \in \mathrm{Dom}(g_1) \cap \ldots \cap \mathrm{Dom}(g_m)$ satisfy $(g_1(\mathbf{p}), \ldots, g_m(\mathbf{p})) \in \mathrm{Dom}(f)$.

↓ to make sure everything is well defined

If

- f is totally differentiable at $(g_1(\mathbf{p}), \ldots, g_m(\mathbf{p}))$, and
- $g_1, \ldots, g_m$ are totally differentiable at $\mathbf{p}$,

then $f \circ (g_1, \ldots, g_m)$ is totally differentiable at $\mathbf{p}$, and, for each $i \in \{1, \ldots, n\}$,

$$\frac{\partial}{\partial x_i}[f(g_1(\mathbf{x}), \ldots, g_m(\mathbf{x}))]\big|_{\mathbf{x}=\mathbf{p}} = \sum_{k=1}^{m} \frac{\partial f}{\partial g_k}(g_1(\mathbf{p}), \ldots, g_m(\mathbf{p})) \cdot \frac{\partial g_k}{\partial x_i}(\mathbf{p}).$$

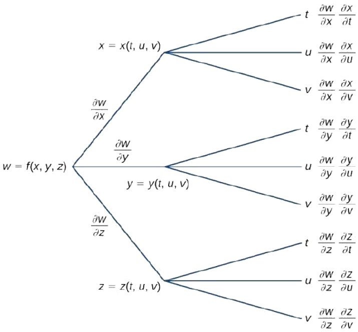
Not tested!

Example

Use the Leibniz Notation to write the Chain Rule for a real variable w dependent upon three real variables, x, y, and z (via the function f), themselves dependent upon three real variables, t, u, and v.

Solution

- Here is a tree diagram for the $(3, 3)$ version of the Multivariable Chain Rule:



to know how many end nodes, just take $m \times n$ in this case is 3×3 which is 9

- This tree diagram yields the following three formulas:

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t},$$
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u},$$
$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}.$$

one for each independent variable

Question: Can tree diagrams for the Multivariable Chain Rule have paths consisting of more than two branches? $\Rightarrow$ not tested

yes, for nested compositions

★ most likely not tested on proofs! ⇒ we say one

Example

- Let f be a real-valued function of two real variables with domain $\mathbb{R}^2$.
- Suppose that f is totally differentiable throughout $\mathbb{R}^2$. → This ensures everything works well.
- Define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) \stackrel{\text{df}}{=} f\left(x^2 - y^2, y^2 - x^2\right)$.

Problem: Show, for all $x, y \in \mathbb{R}$, that $y \cdot g_x(x, y) + x \cdot g_y(x, y) = 0$.

Find g_x and g_y

Solution It is readily verified that g is totally differentiable throughout $\mathbb{R}^2$.

- We can apply the $(2, 2)$ version of the Multivariable Chain Rule to this problem.
- We then have, for all $x, y \in \mathbb{R}$, that

$$
\begin{aligned}
g_x(x, y) &= f_x\left(x^2 - y^2, y^2 - x^2\right) \cdot (2x) + f_y\left(x^2 - y^2, y^2 - x^2\right) \cdot (-2x) \\
&= 2x\left[f_x\left(x^2 - y^2, y^2 - x^2\right) - f_y\left(x^2 - y^2, y^2 - x^2\right)\right], \\
g_y(x, y) &= f_x\left(x^2 - y^2, y^2 - x^2\right) \cdot (-2y) + f_y\left(x^2 - y^2, y^2 - x^2\right) \cdot (2y) \\
&= -2y\left[f_x\left(x^2 - y^2, y^2 - x^2\right) - f_y\left(x^2 - y^2, y^2 - x^2\right)\right],
\end{aligned}
$$

so $y \cdot g_x(x, y) = -x \cdot g_y(x, y)$, which yields $y \cdot g_x(x, y) + x \cdot g_y(x, y) = 0$.

A curve in $\mathbb{R}^2$ is sometimes defined implicitly in the form

circle
$F(x,y) = x^2+y^2-1$ (unit circle)

$$F(x, y) = 0,$$

where F is a real-valued function of two real variables.

How do we know whether such a curve has a unique tangent line at a given point, and if there is a unique tangent line at the given point, then how do we determine it?

Theorem (Existence of a Unique Tangent Line)

- Let F be a real-valued function of two real variables with an open domain.
- Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(F)$ satisfy $F(\mathbf{p}) = 0$.
- Suppose that a curve C in $\mathbb{R}^2$ satisfies $C = \{\mathbf{x} \in \mathrm{Dom}(F) \mid F(\mathbf{x}) = 0\}$.

If

- F is totally differentiable at $\mathbf{p}$, and
- $(F_x(\mathbf{p}), F_y(\mathbf{p})) \neq (0, 0)$,

then there is a unique tangent line to C at $\mathbf{p}$, given by the equation

$$F_x(\mathbf{p}) \cdot (x - a) + F_y(\mathbf{p}) \cdot (y - b) = 0.$$

The existence part of this theorem will be discussed later. not tested!

We prove the uniqueness part right now:

- Let $\mathbf{r}$ be any parametrization of C that satisfies the following properties:
  - $\mathbf{p}$ is not the limit of $\mathbf{r}$ at the endpoints of the parameter space $\mathrm{Dom}(\mathbf{r})$.
  - $\mathbf{r}$ is semi-regular at all parameters whose image is $\mathbf{p}$.
- Let t^* be a parameter whose image is $\mathbf{p}$, i.e., $\mathbf{r}(t^*) = \mathbf{p}$.
- Let g and h denote the first and second component functions of $\mathbf{r}$, respectively.
- Then for all $t \in \mathrm{Dom}(\mathbf{r})$, we have $F(g(t), h(t)) = 0$.
- Using the $(2,1)$ version of the Multivariable Chain Rule, we obtain

$$\begin{aligned} 0 &= F_x(g(t^*), h(t^*)) \cdot g'_\pm(t^*) + F_y(g(t^*), h(t^*)) \cdot h'_\pm(t^*) \\ &= F_x(\mathbf{p}) \cdot g'_\pm(t^*) + F_y(\mathbf{p}) \cdot h'_\pm(t^*) \\ &= (F_x(\mathbf{p}), F_y(\mathbf{p})) \bullet \mathbf{r}'_\pm(t^*). \end{aligned}$$

- Hence, $\mathbf{r}'_\pm(t^*) \perp (F_x(\mathbf{p}), F_y(\mathbf{p}))$, so C has a unique tangent line at $\mathbf{p}$.
- Finally, a point $(x, y) \in \mathbb{R}^2$ is on C if and only if

$$(F_x(\mathbf{p}), F_y(\mathbf{p})) \bullet (x - a, y - b) = 0,$$
$$F_x(\mathbf{p}) \cdot (x - a) + F_y(\mathbf{p}) \cdot (y - b) = 0.$$

Example

Show that the unit circle has a unique tangent line at each of its points.

Solution

- Define $F : \mathbb{R}^2 \to \mathbb{R}$ by $F(x, y) \stackrel{\text{df}}{=} x^2 + y^2 - 1$.
- The unit circle C is then the zero set of F, i.e., $C = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; F(\mathbf{x}) = 0 \right\}$.
- For all $x, y \in \mathbb{R}$, we have $F_x(x, y) = 2x$ and $F_y(x, y) = 2y$.
- F_x and F_y are defined and continuous throughout $\mathbb{R}^2$, so F is totally differentiable throughout $\mathbb{R}^2$.
- Given $\mathbf{p} = (a, b) \in C$, we have $(F_x(\mathbf{p}), F_y(\mathbf{p})) = (2a, 2b) \neq (0, 0)$. $(a,b) \neq (0,0)$
- Hence, C has a unique tangent line at $\mathbf{p}$, given by the equation

$$\begin{aligned} 2a(x - a) + 2b(y - b) &= 0, \\ ax + by &= a^2 + b^2, \\ ax + by &= 1. \end{aligned}$$

- As $\mathbf{p}$ is an arbitrary point on the unit circle, we conclude that the unit circle has a unique tangent line at each of its points.

Similarly, a surface in $\mathbb{R}^3$ is sometimes defined implicitly in the form

$$\boxed{F(x, y, z) = 0,}$$

$x^2+y^2+z^2-1$
↳ sphere

where F is a real-valued function of three real variables.

Theorem (Existence of a Unique Tangent Plane)

- Let F be a real-valued function of three real variables with an open domain.
- Let $\mathbf{p} = (a, b, c) \in \mathrm{Dom}(F)$ satisfy $F(\mathbf{p}) = 0$.
- Suppose that a surface S in $\mathbb{R}^3$ satisfies $S = \{\mathbf{x} \in \mathrm{Dom}(F) \mid F(\mathbf{x}) = 0\}$.

If

- F is totally differentiable at $\mathbf{p}$, and
- $(F_x(\mathbf{p}), F_y(\mathbf{p}), F_z(\mathbf{p})) \neq (0, 0, 0)$,

check that F_x, F_y, F_z are continuous

then there is a unique tangent plane to S at $\mathbf{p}$, given by the equation

$$\underbrace{\boxed{F_x(\mathbf{p}) \cdot (x - a) + F_y(\mathbf{p}) \cdot (y - b)} + F_z(\mathbf{p}) \cdot (z - c)}_{\text{same as previous}} = 0.$$

Let F be a real-valued function of three real variables.

Sphere

Let S denote the zero set of F, i.e., $S \stackrel{\text{df}}{=} \{(x, y, z) \in \mathrm{Dom}(F) \mid F(x, y, z) = 0\}$.

Question: Can we solve the equation $F(x, y, z) = 0$ for z in terms of x and y?

If we can do this, then we have a parametrization of S in terms of two parameters only.

To do this for any F is, of course, wishful thinking.

However, we have a theorem saying that if conditions are right, then this is possible.

Theorem (the Implicit Function Theorem)

- Let F be a real-valued function of three real variables with an open domain.
- Suppose that F is continuously totally differentiable throughout $\mathrm{Dom}(f)$.
- Let $\mathbf{p} = (a, b, c) \in \mathrm{Dom}(F)$ satisfy $F(\mathbf{p}) = 0$.

f_x, f_y, f_z continuous

If $F_z(\mathbf{p}) \neq 0$, then we can find

- a sufficiently small open disk D in $\mathbb{R}^2$ centered at (a, b),
- a sufficiently small open interval I in $\mathbb{R}$ centered at c, and
- a function $z : D \to I$ that is continuously totally differentiable throughout D,

such that $z(a, b) = c$ and, for $(x, y) \in D$, the following statements hold:

- $(x, y, z(x, y)) \in \mathrm{Dom}(F)$ and $F(x, y, z(x, y)) = 0$.
- If $z^* \in I$ satisfies $(x, y, z^*) \in \mathrm{Dom}(F)$ and $F(x, y, z^*) = 0$, then $z^* = z(x, y)$.
- $F_z(x, y, z(x, y)) \neq 0$.

This theorem is not easy to prove, so a proof will not be shown.

z solves $F(x,y,z) = 0$ in terms of x, y.

The Implicit Function Theorem is exactly what justifies implicit differentiation.

Let us put ourselves in the setting of the Implicit Function Theorem.

Given $(x, y) \in D$, we have $F(x, y, z(x, y)) = 0$; we can then apply the $(3, 2)$ version of the Multivariable Chain Rule to get

see slide 30

$$\begin{cases} 0 = F_x(x, y, z(x, y)) \cdot 1 + F_y(x, y, z(x, y)) \cdot 0 + F_z(x, y, z(x, y)) \cdot z_x(x, y), \\ 0 = F_x(x, y, z(x, y)) \cdot 0 + F_y(x, y, z(x, y)) \cdot 1 + F_z(x, y, z(x, y)) \cdot z_y(x, y), \end{cases}$$

$\Rightarrow$ y and x are independent variables

which yield the famous formulas

impt formula!

$$\boxed{z_x(x, y) = -\frac{F_x(x, y, z(x, y))}{F_z(x, y, z(x, y))}} \quad \text{and} \quad \boxed{z_y(x, y) = -\frac{F_y(x, y, z(x, y))}{F_z(x, y, z(x, y))}}.$$

$$F(x,y,z)=0 \;\Big|\; \frac{\partial F}{\partial x} = \frac{\partial F}{\partial x} \cdot \underbrace{\frac{\partial x}{\partial x}}_{=1} + \frac{\partial F}{\partial y} \cdot \underbrace{\frac{\partial y}{\partial x}}_{=0} + \frac{\partial F}{\partial z} \cdot \frac{\partial z}{\partial x} \Rightarrow \text{wrt } x$$

Example

- Define $F : \mathbb{R}^3 \to \mathbb{R}$ by $F(x, y, z) \overset{\text{df}}{=} x^3 + y^3 + z^3 + 6xyz - 1$.
  Note: F is continuously totally differentiable throughout $\mathbb{R}^3$.
- Let $\mathbf{p} = (a, b, c) \in \mathbb{R}^3$ satisfy $F(\mathbf{p}) = 0$.
- Suppose that $F_z(\mathbf{p}) \neq 0$.
- Let z be a real-valued function of two real variables, defined locally around (a, b), that satisfies the conclusions of the Implicit Function Theorem.

Problem: Evaluate $\dfrac{\partial z}{\partial x}(a, b)$.

Solution

- Given $(x, y, z) \in \mathbb{R}^3$, we have

$$F_x(x, y, z) = 3x^2 + 6yz \quad \text{and} \quad F_z(x, y, z) = 3z^2 + 6xy.$$

- Therefore,

$$\checkmark \; \frac{\partial z}{\partial x}(a, b) = -\frac{F_x(a, b, c)}{F_z(a, b, c)} = -\frac{3a^2 + 6bc}{3c^2 + 6ab} = -\frac{a^2 + 2bc}{c^2 + 2ab}.$$

$z_x \Rightarrow$ from previous slide

Let f be a real-valued function of two real variables with an open domain.

Let $\mathbf{p} = (a, b) \in \mathrm{Dom}(f)$.

Recall that the x- and y-partial derivatives of f at $\mathbf{p}$ are defined by differentiating f at $\mathbf{p}$ with respect to the positive x-direction $\mathbf{i}$ and the positive y-direction $\mathbf{j}$, respectively:

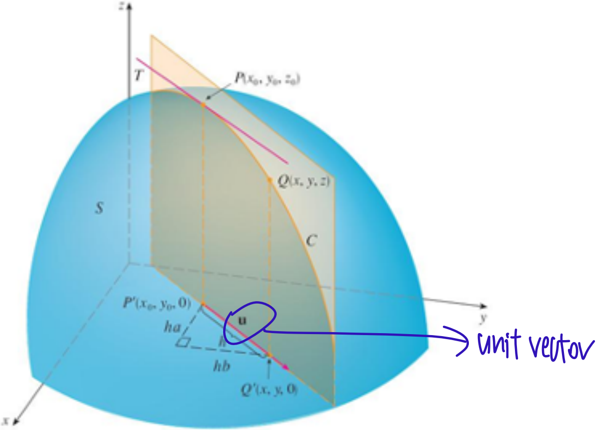
$\rightarrow$ fix $y=b$

$\vec{p} + t \cdot \vec{i} = (a,b) + t \cdot (1,0)$
$= (a+t, b)$

$$
\begin{aligned}
f_x(\mathbf{p}) &\stackrel{\text{df}}{=} \lim_{t\to 0} \frac{f(a+t, b) - f(a,b)}{t} \\
&= \lim_{t\to 0} \frac{f((a,b) + t\cdot(1,0)) - f(a,b)}{t} = \lim_{t\to 0} \frac{f(\mathbf{p} + t\cdot \mathbf{i}) - f(\mathbf{p})}{t}, \\
f_y(\mathbf{p}) &\stackrel{\text{df}}{=} \lim_{t\to 0} \frac{f(a, b+t) - f(a,b)}{t} \\
&= \lim_{t\to 0} \frac{f((a,b) + t\cdot(0,1)) - f(a,b)}{t} = \lim_{t\to 0} \frac{f(\mathbf{p} + t\cdot \mathbf{j}) - f(\mathbf{p})}{t}.
\end{aligned}
$$

If we differentiate f at $\mathbf{p}$ with respect to another direction represented by a unit vector $\mathbf{u}$, then we obtain the directional derivative of f at $\mathbf{p}$ in the direction of $\mathbf{u}$.

Directional Derivatives in Two Dimensions — a Visualization



Directional Derivatives in Two Dimensions — a Formal Definition

Definition (Directional Derivatives in Two Dimensions)

- Let f be a real-valued function of two real variables with an open domain.
- Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^2$.

The directional derivative of f in the direction of $\mathbf{u}$, denoted by "$\mathrm{D}_{\mathbf{u}} f$", is defined as the function with these properties:

- $\mathrm{Dom}(\mathrm{D}_{\mathbf{u}} f) = \left\{ \mathbf{p} \in \mathrm{Dom}(f) \;\middle|\; \lim_{t \to 0} \frac{f(\mathbf{p} + t \cdot \mathbf{u}) - f(\mathbf{p})}{t} \text{ exists} \right\}$.
- For all $\mathbf{p} \in \mathrm{Dom}(\mathrm{D}_{\mathbf{u}} f)$, we have $(\mathrm{D}_{\mathbf{u}} f)(\mathbf{p}) \stackrel{\text{df}}{=} \boxed{\lim_{t \to 0} \frac{f(\mathbf{p} + t \cdot \mathbf{u}) - f(\mathbf{p})}{t}}$.

  Note: For each $\mathbf{p} \in \mathrm{Dom}(\mathrm{D}_{\mathbf{u}} f)$, we call $(\mathrm{D}_{\mathbf{u}} f)(\mathbf{p})$ the "directional derivative of f at $\mathbf{p}$ in the direction of $\mathbf{u}$".

direction

Remark: If the x- and y-partial derivatives of f at $\mathbf{p}$ exist, then observe that

$$f_x(\mathbf{p}) = (\mathrm{D}_{\mathbf{i}} f)(\mathbf{p}) \quad \text{and} \quad f_y(\mathbf{p}) = (\mathrm{D}_{\mathbf{j}} f)(\mathbf{p}).$$

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) \overset{\text{df}}{=} \begin{cases} \dfrac{x^3}{x^2+y^2}, & \text{if } (x,y) \neq (0,0); \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

Problem: Show that the directional derivative of f at $(0,0)$ in any direction exists.

Solution

- Let $\mathbf{u} = (v, w)$ be a unit vector in $\mathbb{R}^2$ (so $v^2 + w^2 = \|\mathbf{u}\|_2^2 = 1$).
- Then for all $t \in \mathbb{R} \setminus \{0\}$, we have

$$\frac{f((0,0) + t \cdot \mathbf{u}) - f(0,0)}{t} = \frac{f(tv, tw)}{t} = \frac{\left[\dfrac{t^3 v^3}{t^2(v^2+w^2)}\right]}{t} = v^3.$$

- Hence, $\displaystyle\lim_{t \to 0} \frac{f((0,0) + t \cdot \mathbf{u}) - f(0,0)}{t} = v^3$.
- As $\mathbf{u}$ is an arbitrary unit vector in $\mathbb{R}^2$, we conclude that the directional derivative of f at $(0,0)$ in any direction exists.

Directional Derivatives in Two Dimensions — Existence

we never go through

Theorem (Total Differentiability Implies the Existence of Directional Derivatives)

- Let f be a real-valued function of two real variables with an open domain.
- Suppose that f is totally differentiable at $\mathbf{p} \in \mathrm{Dom}(f)$.

Then for any unit vector $\mathbf{u}$ in $\mathbb{R}^2$, the directional derivative of f at $\mathbf{p}$ in the direction of $\mathbf{u}$ exists; in fact, $(\mathrm{D}_{\mathbf{u}} f)(\mathbf{p}) = (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet \mathbf{u}$.

Proof

- Some proofs of the theorem use the Multivariable Chain Rule, but we will prove it using the formal definition of total differentiability.
- Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^2$.
- Recall the tangent plane function $L_{f,\mathbf{p}}$ of f at $\mathbf{p}$, which is a linear approximation of f at $\mathbf{p}$ because f is totally differentiable at $\mathbf{p}$.
- For all $t \in \mathbb{R} \setminus \{0\}$, we have

$$\frac{f(\mathbf{p} + t \cdot \mathbf{u}) - f(\mathbf{p})}{t}$$
$$= \frac{f(\mathbf{p} + t \cdot \mathbf{u}) - L_{f,\mathbf{p}}(\mathbf{p} + t \cdot \mathbf{u})}{t} + \frac{L_{f,\mathbf{p}}(\mathbf{p} + t \cdot \mathbf{u}) - L_{f,\mathbf{p}}(\mathbf{p})}{t} + \frac{L_{f,\mathbf{p}}(\mathbf{p}) - f(\mathbf{p})}{t}.$$

- The third summand is 0 because $L_{f,\mathbf{p}}(\mathbf{p}) = f(\mathbf{p})$.
- Let us simplify the second summand:

$$\frac{L_{f,\mathbf{p}}(\mathbf{p}+t\cdot\mathbf{u}) - L_{f,\mathbf{p}}(\mathbf{p})}{t} = \frac{[f(\mathbf{p}) + (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet ((\mathbf{p}+t\cdot\mathbf{u}) - \mathbf{p})] - f(\mathbf{p})}{t}$$
$$= (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet \mathbf{u}.$$

- The absolute value of the first summand can be rewritten as

$$\left|\frac{f(\mathbf{p}+t\cdot\mathbf{u}) - L_{f,\mathbf{p}}(\mathbf{p}+t\cdot\mathbf{u})}{t}\right| = \frac{|f(\mathbf{p}+t\cdot\mathbf{u}) - L_{f,\mathbf{p}}(\mathbf{p}+t\cdot\mathbf{u})|}{\|(\mathbf{p}+t\cdot\mathbf{u}) - \mathbf{p}\|_2},$$

  which $\to 0$ as $t \to 0$ because it is the restriction to the line passing through $(0,0)$ and parallel to $\mathbf{u}$ of the limit

$$\lim_{\mathbf{x}\to\mathbf{p}} \frac{|f(\mathbf{x}) - L_{f,\mathbf{p}}(\mathbf{x})|}{\|\mathbf{x}-\mathbf{p}\|_2} = 0.$$

- Hence, $\displaystyle\lim_{t\to 0} \frac{f(\mathbf{p}+t\cdot\mathbf{u}) - f(\mathbf{p})}{t} = (f_x(\mathbf{p}), f_y(\mathbf{p})) \bullet \mathbf{u}$.
- Finally, as $\mathbf{u}$ is an arbitrary unit vector in $\mathbb{R}^2$, we are done. $\square$

We keep seeing the expression $(f_x(\mathbf{p}), f_y(\mathbf{p}))$, so let us give it a special name.

Definition (the Gradient of a Real-Valued Function of Two Real Variables)

Let f be a real-valued function of two real variables with an open domain.

The gradient of f, denoted by "∇f", is defined as the function with these properties:

- $\mathrm{Dom}(\nabla f) = \{\mathbf{p} \in \mathrm{Dom}(f) \mid \text{The } x\text{- and } y\text{-partial derivatives of } f \text{ at } \mathbf{p} \text{ exist}\}$.
- For all $\mathbf{p} \in \mathrm{Dom}(\nabla f)$, we have $(\nabla f)(\mathbf{p}) \stackrel{\mathrm{df}}{=} (f_x(\mathbf{p}), f_y(\mathbf{p}))$.
  Note: For each $\mathbf{p} \in \mathrm{Dom}(\nabla f)$, we call $(\nabla f)(\mathbf{p})$ the "gradient vector of f at $\mathbf{p}$".

We can thus say that if f and $\mathbf{p}$ satisfy the hypotheses of the previous theorem, then

$$(\mathrm{D}_{\mathbf{u}} f)(\mathbf{p}) = (\nabla f)(\mathbf{p}) \bullet \mathbf{u}.$$

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x, y) \overset{\text{df}}{=} x^2y^3 - 4y$.

(2,5)

Problem: Find the directional derivative of f at $(2, -1)$ in the direction of $\underline{2 \cdot \mathbf{i} + 5 \cdot \mathbf{j}}$.

not unit vector!

Solution

- For all $x, y \in \mathbb{R}$, we have $f_x(x, y) = 2xy^3$ and $f_y(x, y) = 3x^2y^2 - 4$.
- As f_x and f_y are defined and continuous throughout $\mathbb{R}^2$, it follows that f is totally differentiable throughout $\mathbb{R}^2$.

  need a unit vector!
- Now, $(\nabla f)(2, -1) = (-4, 8)$, and the unit vector in the direction of $2 \cdot \mathbf{i} + 5 \cdot \mathbf{j}$ is

$$\mathbf{u} \overset{\text{df}}{=} \frac{2}{\sqrt{29}} \cdot \mathbf{i} + \frac{5}{\sqrt{29}} \cdot \mathbf{j}.$$

- Therefore, $(\mathrm{D}_{\mathbf{u}} f)(2, -1) = (-4, 8) \bullet \left(\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right) = \frac{32}{\sqrt{29}}$.

Directional Derivatives in Two Dimensions — Example 3

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) \stackrel{\text{df}}{=} \begin{cases} \dfrac{x^3}{x^2+y^2}, & \text{if } (x,y) \neq (0,0); \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

Problem: Show that f is not totally differentiable at $(0,0)$.

Solution

- We already encountered this function in Example 1 of this section.
- In our work for that example, we established, for any unit vector $\mathbf{u} = (v,w)$ in $\mathbb{R}^2$, that $\underline{(\mathrm{D}_{\mathbf{u}}f)(0,0) = v^3}$.
- In particular, $f_x(0,0) = (\mathrm{D}_{\mathbf{i}}f)(0,0) = 1$ and $f_y(0,0) = (\mathrm{D}_{\mathbf{j}}f)(0,0) = 0$. $\vec{u} = i = (1,0)$
  $j = (0,1)$
- Hence, $(\nabla f)(0,0) = (1,0)$.
- If f were totally differentiable at $(0,0)$, then, for any unit vector $\mathbf{u} = (v,w)$ in $\mathbb{R}^2$,

$$(\mathrm{D}_{\mathbf{u}}f)(0,0) = \underbrace{(\nabla f)(0,0)}_{=(1,0)} \bullet (v,w) = (1,0) \bullet (v,w) = v.$$

$v^3 \neq v$

- This contradicts the second bullet point.

Nothing stops us from generalizing the concept of the gradient to real-valued functions of several real variables, so we shall make the following definition.

Definition (the Gradient of a Real-Valued Function of Several Real Variables)

- Let $m \in \mathbb{N}$.
- Let f be a real-valued function of m real variables with an open domain.

The gradient of f, denoted by "∇f", is defined as the function with these properties:

- $\mathrm{Dom}(\nabla f) = \{\mathbf{p} \in \mathrm{Dom}(f) \mid \text{All } m \text{ partial derivatives of } f \text{ at } \mathbf{p} \text{ exist}\}$.
- For all $\mathbf{p} \in \mathrm{Dom}(\nabla f)$, we have $(\nabla f)(\mathbf{p}) \overset{\mathrm{df}}{=} (f_{x_1}(\mathbf{p}), \ldots, f_{x_m}(\mathbf{p}))$.
  Note: For each $\mathbf{p} \in \mathrm{Dom}(\nabla f)$, we call $(\nabla f)(\mathbf{p})$ the "gradient vector of f at $\mathbf{p}$".